How To Web Dev Tutorial

Build a Responsive Portfolio Website using HTML & CSS & JavaScript

html

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Portfolio Website</title>
    <!-- Custom CSS -->
    <link rel="stylesheet" href="style.css" />
    <!-- Boxicons CDN -->
    <link href="https://unpkg.com/boxicons@2.1.4/css/boxicons.min.css" rel="stylesheet">
  </head>
  <body>
    <!--Navigation bar-->
    <nav>
        <!--<span></span> gives the Logo text a gradiant effect.-->
        <!--This will be set to the left.-->
        <a href="#" class="logo"><span>Linda</span></a>
        <!--ul = unordered list-->
        <!--This will be centered-->
        <ul class="links">
            <a href="#home">Home</a>
            <a href="#about">About</a>
            <a href="#services">Services</a>
            <a href="#contact">Contact</a>
        </ul>

        <!--Contact button to the right-->
        <i class="bx bx-menu" id="menu"></i>

        <!--CTA Button-->
        <a href="#contact" class="btn">Contact Me</a>

    </nav> <!--This sets the navigation bar-->
    <!--Navigation bar-->

    <!-- Home Section -->
  <section id="home">
    <img src="image.jpg" alt="">

    <div class="info-box">
      <h1><span>Hi, I'm Linda</span></h1>
      <h3><span>Web Developer</span> based in Toronto</h3>
      <p>Passionate about building responsive, modern, and user-friendly web applications.</p>
    </div>

    <div class="btn-box">
      <div class="btn">Hire</div>
      <div class="btn">Contact Me</div>
```

```
    </div>
  </section>
    <!--Home Section-->

  <!--About me section-->
  <section id="about">
    <div class="about-box">
      <div class="about-info">
        <h1><span>About Me</span></h1>
        <p>I am a front end developer focused on creating smooth visual experiences and
responsive interfaces</p>
      </div>
  <!--About me section-->
      <!--The tutorial uses a Boxicon "<i class= 'bx bx-chevron-right' ></i>" that appears to be
unavailable in the free version. Rather than spending additional time searching for a
replacement, I substituted a dash because the icon is purely cosmetic and does not affect
functionality..-->
      <!--Skills section-->
      <h2><span>Skills</span></h2>
      <div class="skills">
        <ul>
          <li>
            <span>-HTML</span>
          </li>
          <li>
            <span>-CSS</span>
          </li>
          <li>
            <span>-JavaScript</span>
          </li>
        </ul>
        <ul>
          <li>
            <span>-React</span>
          </li>
          <li>
            <span>-Node</span>
          </li>
          <li>
            <span>-MySQL</span>
          </li>
        </ul>
      </div>
    </div>
    <img src="image.jpg" alt="">
  </section>
  <!--Skills section-->
  <!--Services section-->
```

```
  <section id="services">
    <h1 class="header"><span>Services</span></h1>
      <div class="services-container">
        <div class="box">
          <h1><span>Web Design</span></h1>
          <p>Passionate about building responsive, modern, and user-friendly web
applications</p>
          <img src="web-dev.jpg" alt="">
        </div>
        <div class="box">
          <h1><span>Web Development</span></h1>
          <p>Passionate about building responsive, modern, and user-friendly web
applications</p>
          <img src="web-design.jpg" alt="">
        </div>
        <div class="box">
          <h1><span>Graphics Design</span></h1>
          <p>Passionate about building responsive, modern, and user-friendly web
applications</p>
          <img src="graphic.jpg" alt="">
        </div>
      </div>
  </section>
  <!--Services section-->

  <!--Video stopped at 27 minutes
https://www.youtube.com/watch?v=GnFH9Eg-OuM&list=PL4FL-XepWXttYviyLh0yEiNOhAj_qCY
s4&index=3-->

  <!--Contact section-->
  <!--section#contact makes a section with the id of contact-->
  <section id="contact">
    <!--h1.header short hand learned-->
    <h1 class="header"><span>Contact</span></h1>
    <form action="#">
      <!--.input-box-->
      <div class="input-box">
        <span>Full Name</span>
        <input type="text" placeholder="Enter your full name">
      </div>
      <div class="input-box">
        <span>Full Email</span>
        <input type="text" placeholder="Enter your full email">
      </div>
      <div class="input-box">
        <span>Message</span>
        <input type="text" placeholder="Enter your message">
      </div>
```

```
      <!--Outside the div but inside the form-->
      <!--.btn shorthand for making a button-->
      <div class="btn">Submit</div>
    </form>
    <!--Contact section-->
  </section>
  <!--Footer-->
  <footer>
  <!--col-left this just makes a column to the left <col-left></col-left>, .col-left - this makes a
column div class <div class="col-left"></div>-->
  <div class="col-left">
    <!--Not sure if it is me or my skill, or just not knowing how boxicon works I can't get them to
work.-->
    <div class="col-box">
      <span>Address <i class='bx bxs-location-plus'></i></span>
      <span>1234 Maple Street Springfiedl Il 62701</span>
    </div>
    <div class="col-box">
      <span>Phone <i class='bx bxs-phone'></i></span>
      <span>1234567890</span>
    </div>
    <div class="col-box">
      <span>Email <i class='bx bxs-envelope'></i></span>
      <span>sampleLinda@gmail.com</span>
    </div>
  </div>

  <div class="col-right">
    <span>About My Service</span>
    <p>Explanation goes here</p>
    <div class="social-icons">
      <a href="#" target="_blank"><i class='bx bxl-github'></i></a>
      <a href="#" target="_blank"><i class='bx bxl-linkedin'></i></a>
      <a href="#" target="_blank"><i class='bx bxl-twitter'></i></a>
      <a href="#" target="_blank"><i class='bx bxl-facebook'></i></a>
    </div>
  </div>
  </footer>
  <!--Footer-->

  <!--In his video he has js named script, I am used to app.js so I named it accordingly.-->
    <script src="app.js"></script>
  </body>
</html>
```

CSS

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  text-decoration: none;
  list-style: none;
  scroll-behavior: smooth;
  font-family: "Poppins";
}
:root {
  --background: rgb(24, 24, 24);
  --second-background: rgb(21, 21, 21);
}
body {
    min-height: 100vh;
    background-color: var(--background);
}
nav {
    position: fixed;
    top: 0;
    width: 100%;
    padding: 2em 15%; /* 2em at the top and bottom, 15% on left and right */
    display: flex;
    align-items: center;
    justify-content: space-between;
    background-color: rgba(15, 15, 15, 0.8);
    backdrop-filter: blur(10px);
    z-index: 1000; /* Ensures the bar stays on top of everything else */
}/*Sets the nav bar to be fixed at the top and span the whole width of the screen.*/
ul {
    display: flex;
    align-items: center;
    gap: 3em;

}
nav ul {
    color: white;
    opacity: 0.7; /*For the hover effect*/
    transition: 0.2s ease-in-out; /*Transition for the hover effect*/

}
nav ul a:hover{
    opacity: 1; /*Brings the links being hovered over to solid*/

}
.logo {
    font-size: 2em;
    color: white;
```

```
    font-weight: 800;
    opacity: 0.8;
    transition: 0.2s ease-in-out;

}
.logo:hover {
    opacity: 1;

}
.btn {
    padding: 0.5em 1em;
    color: white;
    font-weight: 500;
    letter-spacing: 1px;
    border-radius: 0.5em;
    font-size: 1.1em;
    border: 2px solid white;
    cursor: pointer
}
.links a {
    color: white;
    opacity: 0.7;
    transition: 0.2s ease-in-out;
}

.links a:hover {
    opacity: 1;
}
/*Stopped at 9:30
https://www.youtube.com/watch?v=GnFH9Eg-OuM&list=PL4FL-XepWXttYviyLh0yEiNOhAj_qCY
s4&index=2*/

.btn {
    padding: 0.5em;
    color: white;
    font-weight:500;
    letter-spacing: 1px;
    border-radius: 0.5em;
    font-size: 1.1em;
    border: 2px solid white;
    cursor: pointer; /*Indicates that the button is clickable when hovering over the button.*/
    transition: 0.3s ease-in-out;
}
.btn:hover{
    background-color: white;
    color:var(--background);
}
/*Setting the <span></span> setting to create the gradiant text effect*/
```

```
/*This is a linear gradiant function = linear-gradiant being set to go to the right.*/
span{
    background: linear-gradient(to right, #6cff52, #3ec5ff);
    background-clip: text; /*To clip the text to the background*/
    color: transparent; /*Makes the text transparent*/

}
#menu{
    color:white;
    font-size:3em;
    display: none;
}

/*Styling of the home section*/
section{
    min-height: 100vh;
    padding: 5% 15%; /*5% padding on top and bottom, 15% padding on the sides*/

}
#home{
    display: flex;
    flex-direction: column; /*Makes it where all of the home section centers and stacks on each
other, the image followed by the text*/
    align-items: center;
    justify-content: center;
    gap: 2em;
    color: white; /*Why not have a global color for text color have it be white and then insert the
var(--text color) he already had us do var inserts*/

}
/*Modifying the image size*/
#home img{
    width: 18vw; /*18% of the view port width*/
    border-radius: 50%; /*Makes the image circular when it is by default a square.*/
    margin-top: 3em;
}
.info-box{
    display: flex;
    flex-direction: column;/*Brings the text ontop of each other.*/
    text-align: center;
    gap: 0.5em;
    max-width: 600px;/*Aids in the responsiveness as well*/
}
.info-box h1{
    font-size: 5em;
    font-weight: 500;

}
```

```
.info-box h3{
    font-size: 2em;
    font-weight:400;
}
.info-box p{
    opacity: 0.7;
}
.btn-box{
    display: flex;
    gap: 1em;
}
.btn-box .btn:nth-of-type(2){
    background-color: white;
    color: var(--background);
}
.btn-box .btn:nth-of-type(2):hover{
    background-color: var(--background);
    color: white;
}
/*Styling of the home section*/

/*Styling the scroll bar*/
::-webkit-scrollbar{
    width:20px;
    background-color: var(--background);

}
::-webkit-scrollbar-thumb{
    background: linear-gradient(to bottom, #6cff52, #3ec5ff);
}
/*Styling the scroll bar*/

/*Styling for the About section*/
#about{
    background-color: var(--background);
    display:flex;
    align-items:center;
    justify-content:center;
    gap:5em;
}
#about img{
    width: 30vw;
    border-radius: 1em;
}/*Modifies the image in the bottom of the about section.*/
.about{
    display:flex;
    flex-direction: column;
    text-align:left;
```

```
    gap: 2em;
}
.about-box h1{
    font-size: 5em;
}
.about-box p{
    color: white;
    opacity:0.8;
}
/*Styling for the About section*/

/*SKills*/
.skills{
    display: flex;
    text-align: left;
    gap: 10em;
}
.skills ul{
    font-size: 1.3em;
    font-weight: 500;
    display: flex;
    flex-direction: column;
    color: white;
    opacity: 0.8;
    align-items: baseline; /*Perfectly align the items on the left.*/
}
.skill ul li span{
    display: flex;
    align-items: center;
    gap: 0.5em;
    font-weight: 1.3em;
    transition: 0.3 ease-in-out;
} /*Give space between the text and the boxicon.*/
.skills ul li:hover{
    transform: translatex(10px); /*Hovering over the skills causes it to shift over to the right a
little.*/

}
.skills ul li{
    font-weight: 1.2em;
}
.about-box h2{
    font-size: 2.5em
}
/*SKills*/
/*Services*/
#services{
    background-color: var(--background);
```

```
}
.header{
    align-content: center;
    font-size: 5em;
    margin: 0.5em 0;

}
.services-container{
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
    gap: 2em;

}
.box{
    border: 3px solid rgb(49,49,49);
    border-radius:0.5em;
    display: flex;
    flex-direction: column;
    justify-content: center;
    overflow: hidden;
    object-fit:cover;
    padding: 3em;
    gap: 2em;
    color: white;
    cursor: pointer;
    transition: 0.3s ease-in-out;

}
.box:hover{
    border: 3px solid rgb(100,100,100);

}
.box p{
    opacity: 0.8;

}
.box h1{
    font-size: 2em;

}
/*.box img{
    min-width: 25px;
    border-radius: 0.5em;
}*/
.box img {
    width: 100%;
    height: auto;
    border-radius: 0.5em;
```

```
}
/*Encountering a bug the tiles are not columnizing and zooming out to 20% mimics how they appear in the tutorial.*/
/*Services*/

/*Contact*/
#contact {
    background-color: var(--second-background);

}
form {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    gap: 2em;

}
form .input-box {
    display: flex;
    flex-direction: column;
    align-items: baseline; /*Keeps it default aligned to the left*/
    gap: 1em;
}
.input-box span{
    font-size: 1.5em;

}
font-input {
    font-size: 1.3em;
    padding: 1em 5em;
    border-radius: 0.5em;
    border: none;
    outline: none;
}
form input:focus {
    background: rgb(229,229,229);
    /*Sets the background to this color when hovering over the text*/
}
/*Contact*/

/*Footer*/
/*
When doing the tutorial I messed up and put a .footer where the YouTube video he did not infact put a period. I asked for an explainer on why I made a mistake as so far all calls to the html started with a .(class itme)
It all comes down to how CSS rules are written to tell the browser **what** on the page you're trying to style:
```

1. **Tag Selectors (No Dot):** When you write `footer`, `div`, `h1`, or `p` without a period, you are telling the browser, *"Find every actual HTML element with this name."* Because your code used the `<footer >` tag, `footer` targets that structural element directly.
2. **Class Selectors (With a Dot):** When you put a period in front of a word like `.footer`, `.col-left`, or `.btn`, the period acts as a prefix that tells CSS, *"Don't look for a tag; look for an element with `class="..."`."*

When CSS saw `.footer` in your stylesheet, it went looking through your HTML for `<div class="footer">` or `<section class="footer">`. Because your HTML used `<footer>` without a `class="footer"` attribute attached to it, the browser couldn't find a match. It simply ignored all those flexbox rules, leaving your content unstyled and stuck on the left!

Deleting that period stripped away the class filter, allowing CSS to apply the styling directly to your HTML `<footer>` tag.
*/

```
footer {
    display: flex;
    align-items:center;
    justify-content: center;
    /*Brings the column left and right into the center*/
    gap: 10em;
    bottom: 0; /*Ensures it stays at the bottom*/
    background-color: var(--background);
    width: 100%;
    padding: 5em 15%;
}
.col-left {
    display: flex;
    flex-direction: column;
    gap: 1em;

}
.col-box {
    display: flex;
    align-items: center;
    gap: 1em;

}
.col-box i{
    font-size: 2em; /*Increases the icon size*/

}
.col-box span {
    font-size: 1.2em;

}
.col-right {
```

```
    display: flex;
    flex-direction: column;
    gap: 1em;
    color: white;
    max-width: 500px;
}
.col-right span {
    font-size: 1.3em;
}
.col-box .social-icons {
    display: flex;
    align-items: baseline;
    justify-content: left;
    gap: 1em;
}
.col-right .social-icons i{
    font-size: 2em;
    transition: 0.2 ease-in-out;
    cursor: pointer;

}
.social-icons i:hover {
    color:white;
    /*Sets the color of the icons when you hover over them*/

}
/*Footer*/
/*Responsiveness settings from JavaScript*/
@media(max-width:968px){
    section{
        padding: 8em 15%;
    }
    nav .btn{
        display:none; /*Deletes the button on the header*/
    }
    #menu{
        display: block;
    }
    .links{
        position: absolute;
        top: 100%;
        right:0;
        width: 100%;
        padding: 1em 3em;
        color: white;
        display: flex;
        flex-direction: column;
        text-align: center;
```

```
        background: rgba(0,0,0,0.9);
        display: none;
    } /*List for the responsive drop down menu*/
    .links a{
        margin: 2em 0;
        display: block;
        font-size: 1.5em;
        color: white;
    }
    .links.active{
        display: block;
    }
    #home h1{
        font-size:4em;
    }
    #home h3{
        font-size: 2em;
    }
    #home img{
        width: 24vw;
    }
    #about{
        flex-direction:column;
    }
    #about img{
        max-width:32vw;
    }
    #about h1{
        font-size: 3em;
    }
    #about ul{
        font-size: 1em;
    }

}
@media(max-width:768px){
    #home h1{
        font-size:2.5em;
    }
    #home h3{
        font-size: 1.5em;
    }
    #home img{
        width: 32vw;
    }
    #about .skills{
        gap: 3em;
    }
```

```
    #services .header{
        font-size: 3em;
    }
    #contact .header{
        font-size: 3em;
    }
    #contact .input-box span{
        font-size: 1.2em;
    }
    .input-box input{
        padding: 0.5em 1em;
    }
    footer{
        flex-direction: column;
        gap: 5em;
        align-items: baseline;
    } /*Even though they are set to column left and right , when the view port hist this number they
become a column*/
}
/*Responsiveness settings from JavaScript*/
```

javascript

```
/*
-----------------------------------------------------
Project:
Dynamic GitHub Portfolio

Purpose:
Retrieve my public GitHub repositories and
display them dynamically within my portfolio.

Tutorial:
https://www.youtube.com/watch?v=GnFH9Eg-OuM&list=PL4FL-XepWXttYviyLh0yEiNOhAj_qCYs4&index=2

Personal Modifications:
- Corrected GitHub API endpoint.
- Improved comments.
- Added error handling explanation.
- Documented DOM interaction.
-----------------------------------------------------
*/

/*Drop down menu */
const menu = document.querySelector('#menu');
const nav = document.querySelector('.links');
menu.onclick = () => {
  menu.classList.toggle('bx-x');
  nav.classList.toggle('active');
}

/*Drop down menu */
```

Footer template

HTML version 1

```
<!--Footer-->
  <footer>
  <!--col-left this just makes a column to the left <col-left></col-left>, .col-left - this makes a column div class <div class="col-left"></div>-->
  <div class="col-left">
    <!--Not sure if it is me or my skill, or just not knowing how boxicon works I can't get them to work.-->
    <span>Address</span>
    <span>1234 Maple Street Springfiedl Il 62701</span>
  </div>
  <div class="col-left">
    <span>Phone</span>
    <span>1234567890</span>
  </div>
  <div class="col-left">
    <span>Email</span>
    <span>sampleLinda@gmail.com</span>
  </div>

  <div class="col-right">
    <span>About My Service</span>
    <p>Explanation goes here</p>
    <div class="social-icons">
      <a href="#" target="_blank"><i class='bx bxl-github'></i></a>
      <a href="#" target="_blank"><i class='bx bxl-linkedin'></i></a>
      <a href="#" target="_blank"><i class='bx bxl-twitter'></i></a>
      <a href="#" target="_blank"><i class='bx bxl-facebook'></i></a>
    </div>
  </div>
  <!--Footer→
```

HTML version 2 with icons

```
<!--Footer-->
  <footer>
  <!--col-left this just makes a column to the left <col-left></col-left>, .col-left - this makes a column div class <div class="col-left"></div>-->
  <div class="col-left">
    <!--Not sure if it is me or my skill, or just not knowing how boxicon works I can't get them to work.-->
    <div class="col-box">
      <span>Address <i class='bx bxs-location-plus'></i></span>
```

```
      <span>1234 Maple Street Springfiedl Il 62701</span>
    </div>
    <div class="col-box">
      <span>Phone <i class='bx bxs-phone'></i></span>
      <span>1234567890</span>
    </div>
    <div class="col-box">
      <span>Email <i class='bx bxs-envelope'></i></span>
      <span>sampleLinda@gmail.com</span>
    </div>
  </div>

  <div class="col-right">
    <span>About My Service</span>
    <p>Explanation goes here</p>
    <div class="social-icons">
      <a href="#" target="_blank"><i class='bx bxl-github'></i></a>
      <a href="#" target="_blank"><i class='bx bxl-linkedin'></i></a>
      <a href="#" target="_blank"><i class='bx bxl-twitter'></i></a>
      <a href="#" target="_blank"><i class='bx bxl-facebook'></i></a>
    </div>
  </div>
  </footer>
  <!--Footer-->
```

---

css

```
/*Footer*/
footer {
    display: flex;
    align-items:center;
    justify-content: center;
    /*Brings the column left and right into the center*/
    gap: 10em;
    bottom: 0; /*Ensures it stays at the bottom*/
    background-color: var(--background);
    width: 100%;
    padding: 5em 15%;
}
.col-left {
    display: flex;
    flex-direction: column;
    gap: 1em;

}
.col-box {
    display: flex;
```

```
    align-items: center;
    gap: 1em;

}
.col-box i{
    font-size: 2em; /*Increases the icon size*/

}
.col-box span {
    font-size: 1.2em;

}
.col-right {
    display: flex;
    flex-direction: column;
    gap: 1em;
    color: white;
    max-width: 500px;
}
.col-right span {
    font-size: 1.3em;
}
.col-right .social-icons {
    display: flex;
    align-items: baseline;
    justify-content: left;
    gap: 1em;
}
.col-right .social-icons i{
    font-size: 2em;
    transition: 0.2s ease-in-out;
    cursor: pointer;

}
.social-icons i:hover {
    color:white;
    /*Sets the color of the icons when you hover over them*/

}
/*Footer*/
```

ChatGPT Quiz

This repository is where I organize and catalog the programming tutorials I've completed, giving me a place to revisit, expand upon, and continue improving them over time.

Learning by building works best for me. Rather than simply following a tutorial once, I use each project as an opportunity to document what I've learned, experiment with new ideas, debug problems, and implement my own improvements beyond the original scope.

Original tutorial authors are credited throughout this repository. My goal isn't to memorize tutorials—it's to understand how they work well enough to modify, improve, and eventually build similar projects on my own.

I had ChatGPT quiz me on the code I answered and was rated on my level of awareness:
I like these answers because they're **your reasoning**, not textbook definitions. That's exactly how I want to do these reviews.

Here's how I'd grade them.

---

1. Why use `<ul>`?

Your answer:

> Besides being how he decided to code it also allowed for further formatting actions later.

Grade: 8.5/10

You're on the right track.

The main reason is that a navigation menu is **a list of navigation items**. HTML provides `<ul>` specifically for unordered lists, so it is the semantic choice. Then CSS can easily style the entire list with Flexbox, spacing, hover effects, etc.

You recognized the styling benefit, which is an important part of why tutorials do it.

---

2. Why both `class="bx bx-menu"` and `id="menu"`?

Your answer:

> Because bx bx I assume is the boxicon... id="menu" to allow #menu later.

Grade: 10/10

Exactly.

* `class="bx bx-menu"` tells Boxicons **which icon** to display.

* `id="menu"` gives **your JavaScript and CSS** something unique to select.

One is for the icon library.

One is for your code.

---

3. Why use `id="about"`?

Your answer:

> To allow the #about to work when styling and calling it in the css.

Grade: 9.5/10

Correct.

There's one more benefit:

```html
<a href="#about">
```

scrolls directly to

```html
<section id="about">
```

So the ID is used by:

* CSS
* JavaScript
* Anchor navigation

You already recognized two of those.

---

4. Viewport meta tag

Your answer:

> To auto adjust the screen to fill the screen of the display device.

Grade: 10/10

Perfect.

Without it, phones pretend they're much wider than they really are and then shrink the page to fit, which makes responsive layouts behave strangely.

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5. Why put `<script>` at the bottom?

Your answer:

> It calls and loads the javascript... then applies the configurations listed there.

Grade: 8/10

Correct, but there's an important reason **why** it goes at the bottom.

Imagine this:

The browser reads:

```html
<script src="app.js"></script>
```

before it has even reached

```html
<i id="menu">
```

JavaScript asks:

```javascript
document.querySelector("#menu")
```

The browser responds:

> "What menu? I haven't read that part of the page yet."

It returns `null`.

By putting the script at the bottom, the browser has already created all the HTML elements before JavaScript starts looking for them.

Overall

I'd honestly give you about a **9.2/10**.

More importantly, your answers show something I've noticed over the past few weeks.

A month ago, you would usually answer:

> "Because that's what the tutorial did."

Now your answers sound like:

> "This exists so CSS can reference it."

> "This ID lets JavaScript find it."

> "The viewport adjusts to the device."

That's a shift from describing *what* the code is to explaining *why* it's there. That's the kind of understanding that carries over when you start building your own portfolio instead of following a tutorial.